PAP Many Daughters
Research Team ID:

PAP for Hypothesis 1

H1. *Having daughters leads to stronger support for gender equality (as measured by attitudes).*
That is, denoting *Y* as a measure of support for gender equality, (*Y* for respondents with daughters) > Y for respondents without daughters).

Formally, letting

- $Y_i = \mathbf{gender_attitudes_i}$ index for person i (the sum of the items plh0395i01…plh0395i06 in SOEP 2019), and
- $D_i = 1$ if person i has biological children and has at least one biological daughter born prior to interview ($\boldsymbol{having_daughter_i}$), 0 otherwise,

the hypothesis is

$$H1\colon E[Y_i|D_i = 1, \boldsymbol{X}_i] > E[Y_i|D_i = 0, \boldsymbol{X}_i],$$

where $\boldsymbol{X}_i$ denotes a set of individual-specific controls described below.

1. Variables

- Key Independent Variable: **having_daughter**
  - Definition: Indicator equal to 1 if the respondent has biological children and at least one biological daughter born before the 2019 interview year; 0 otherwise.
  - Construction:
    - Recode any "not applicable" code (–2) in each kidsex01…kidsex19 to valid non-missing. For each child slot j, check if kidsexj=2 (female) and kidgebj < syear. Set having_daughter = 1 if any child meets those conditions; else 0.
- Dependent Variable: **gender_attitudes**
  - Items (all from 2019):
    - plh0395i01: "It's good that legislators take into account that not everyone can be classified as either male or female."
    - plh0395i02: "It should be accepted in our society that not everyone identifies with their biological sex."
    - plh0395i03: "It's completely natural that some people are born with both male and female biological features."
    - plh0395i04: "Sex assignment is not so important for a person's development."
    - plh0395i05: "Schools and preschools should promote non-gender-typical activities that are not typical for children's gender."
    - plh0395i06: "It should be accepted in our society that same-sex couples want to have children."
  - Coding and aggregation:
    - Original responses run from 1 ("Completely disagree") to 7 ("Fully agree"); negative values (<0) indicate non-response. Recode all negative values to missing. Sum the six items for respondents in 2019 to create gender_attitudes (range: 6–42).
- Control Variables
  - Measured or recoded at the 2019 interview, with any negative codes set to missing where indicated:
    - sumkids: Total number of biological children.

        - age_at_int: Computed as 2019 – gebjahr if gebjahr > 0; otherwise missing.
        - female: Original sex variable (1 = male, 2 = female), renamed to female to state the included category in the regression.
        - erwerb: Employment status collapsed from plb0022_h:
            - Category 1 if plb0022_h=9 (not in labor force)
            - Category 2 if plb0022_h in {3,4,5,6,7,8,10,11} (non-regular employment phases)
            - Category 3 if plb0022_h=2 (part-time)
            - Category 4 if plb0022_h=1 (full-time)
        - pld0131_v2: Marital status; recoded to missing if <0.
        - plh0258_v9: Religious affiliation; recoded to missing if <0 ("keine Angabe").
        - hghinc: Monthly household net income in EUR; recode to negative values to missing.
        - migback: Migration background indicator.
        - sampreg: Geographic region of the SOEP sample (East vs. West Germany).
        - pgpsbil: Highest school-leaving qualification.
    - Optional additional covariates for LASSO Selection
        - To allow a data-driven check for omitted-variable bias, the following variables are included in the LASSO selection step (all recoded so negative values → missing):
            - corigin (respondent's country of birth, ISO numeric code)
            - fprofedu and mprofedu (father's and mother's highest professional education)
            - locchildh (place where respondent was raised up to age 15; 1=large city … 4=countryside)
            - hlf0135 (distance of dwelling to city centre; 1=city-centre flat … 6=≥ 60 km)
            - hlf0151 (level of worry about crime in neighbourhood; 1=not worried … 4=very worried)
            - nums and numb (numbers of sisters and brothers reported by the respondent)

2. Inclusion/exclusion criteria:

We include:

- Survey year = 2019 only (cross-section of the SOEP 2019 wave)
- Respondents living in private households (SOEP pop codes 1 or 2)
- Respondents aged ≥ 18 at interview (computed as 2019 – gebjahr)
- Respondents with at least one biological child (sumkids > 0)
- Observations with nonmissing dependent variable (gender_attitudes) or the key independent variable (having_daughter) expressions
- Observations with nonmissing values on any prespecified control (rows with missing values will be dropped automatically by the regression routine)

3. Exact analysis:

We estimate the following OLS model on the dataset described with the above inclusion and exclusion criteria and variables.

$$
\begin{aligned}
&gender_attitudes_i \\
&\quad = \beta_0 + \beta_1\, having_daughter_i + \beta_2 sumkids_i + \beta_3\, hghinc_i + \beta_4 female_i \\
&\quad + \beta_5 age_at_int_i + \sum_{j=2}^{4} \beta_{6,j} I(erwerb_i = j) + \sum_{j=2}^{5} \beta_{7,j} I(pld0131_v2_i = j) \\
&\quad + \sum_{j=2}^{5} \beta_{8,j} I(plh0258_v9_i = j) + \beta_9 migback_i + \beta_{10} I(sampreg_i = 2) \\
&\quad + \sum_{j=2}^{8} \beta_{11,j} I(pgpsbil_i = j) + \sum_{s=2}^{S} \beta_{12,s} control_sel_{i,s} + \varepsilon_i
\end{aligned}
$$

where $control_sel_{i,s}$ are the S selected variables from our Lasso variable selection approach.

4. Interpretation of results:

We will focus on the estimated coefficient on having_daughter (β_1) and its associated t-statistic (using robust SEs) and p-value.

- **Statistic of interest:** the coefficient estimate for having_daughter, together with its t-statistic and (two-sided) p-value.
- **Interpretation:**
  - If the coefficient is positive and its two-sided p-value is below 0.05, we take that as support "in line with the hypothesis."
  - If the coefficient is zero or negative ("not in line with our hypothesis"), or the one-sided p-value is 0.05 or higher, we conclude there is no support for Hypothesis 1.

PAP for Hypothesis 2

H2. *Having daughters leads to more left-leaning political preferences.*
That is, denoting Y as a measure of left-leaning political preferences,
(Y for respondents with daughters) > (Y for respondents without daughters).

Formally, letting

- $Y_i = \text{plh0004}\mathbf{_v2_i}$ index for person i (inversed version of plh0004 in SOEP), and
- $D_i = 1$ if person i has biological children and has at least one biological daughter born prior to interview ($\boldsymbol{having_daughter_i}$), 0 otherwise,

the hypothesis is

$$H1\text{: } E[Y_i|D_i = 1, \boldsymbol{X}_i] > E[Y_i|D_i = 0, \boldsymbol{X}_i],$$

where $\boldsymbol{X}_i$ denotes a set of individual-specific controls described below.

5. Variables

- Key Independent Variable: **having_daughter**
  - Definition: Indicator equal to 1 if the respondent has biological children and at least one biological daughter born before the respective interview year (2009, 2014, 2019); 0 otherwise.
  - Construction:
    - Recode any "not applicable" code (–2) in each kidsex01…kidsex19 to valid non-missing. For each child slot j, check if kidsexj=2 (female) and kidgebj < syear. Set having_daughter = 1 if any child meets those conditions; else 0.
- Dependent Variable: **plh0004_v2**
  - Original SOEP variable: plh0004 ("Political attitude")
  - Coding and aggregation:
    - Original responses run from 0 ("completely left") to 10 ("completely right"); negative values (<0) indicate non-response. Reverse the scale so 0 represents "completely right" and 10 represents "completely left" (accordingly with values in between). Recode all negative values to missing.
- Control Variables
  - Measured or recoded at the interviews in 2009, 2014, and 2019, respectively, with any negative codes set to missing where indicated:
    - sumkids: Total number of biological children.
    - hghinc: Monthly household net income in EUR; recode to negative values to missing.
    - female: Original sex variable (1 = male, 2 = female), renamed to female to state the included category in the regression.
    - age_at_int: Computed as syear – gebjahr if gebjahr > 0; otherwise missing.
    - erwerb: Employment status collapsed from plb0022_h:
      - Category 1 if plb0022_h=9 (not in labor force)
      - Category 2 if plb0022_h in {3,4,5,6,7,8,10,11} (non-regular employment phases)
      - Category 3 if plb0022_h=2 (part-time)
      - Category 4 if plb0022_h=1 (full-time)
    - pld0131_v2: Marital status; recoded to missing if <0.
    - migback: Migration background indicator.

        - sampreg: Geographic region of the SOEP sample (East vs. West Germany).
        - pgpsbil: Highest school-leaving qualification.
    - Optional additional covariates for LASSO Selection
        - To allow a data-driven check for omitted-variable bias, the following variables are included in the LASSO selection step (all recoded so negative values → missing):
            - corigin (respondent's country of birth, ISO numeric code)
            - fprofedu and mprofedu (father's and mother's highest professional education)
            - locchildh (place where respondent was raised up to age 15; 1=large city … 4=countryside)
            - hlf0135 (distance of dwelling to city centre; 1=city-centre flat … 6=≥ 60 km)
            - hlf0151 (level of worry about crime in neighbourhood; 1=not worried … 4=very worried)
            - nums and numb (numbers of sisters and brothers reported by the respondent)
            - religious affil (Religious affiliation, based on plh0258_v9 and imputed with data from surrounding years of the same variable as well as plh0258_v10 and plh0258_v11, on optional list due to potential loss of data due to sparseness in this variable for the years 2009 and 2014)

6. Inclusion/exclusion criteria:

We include:

- Survey years 2009, 2014, 2019 only (2005 also contains plh0004 but many of our controls are not available for this year, reducing the sample)
- Respondents living in private households (SOEP pop codes 1 or 2)
- Respondents aged ≥ 18 at interview (computed as syear – gebjahr)
- Respondents with at least one biological child (sumkids > 0)
- Observations without missing dependent variable (gender_attitudes) or the key independent variable (having_daughter) expressions
- Observations without missing values on any prespecified control (rows with missing values will be dropped automatically by the regression routine)

7. Exact analysis:

We estimate the following random effects panel regression model (using GLS estimation) on the dataset described with the above inclusion and exclusion criteria and variables.

$$
\begin{aligned}
&household_chore_attitudes_{i,t} \\
&\quad = \beta_0 + \beta_1\, having_daughter_{i,t} + \beta_2 female_{i,t} + \beta_3 sumkids_{i,t} \\
&\quad + \beta_5\, hghinc_{i,t} + \beta_6 age_at_int_{i,t} + \sum_{j=2}^{4} \beta_{7,j} I\left(erwerb_{i,t} = j\right) \\
&\quad + \sum_{j=2}^{4} \beta_{8,j} I\left(pld0131_v2_{i,t} = j\right) + \beta_9 migback_{i,t} + \beta_{10} I\left(sampreg_{i,t} = 2\right) \\
&\quad + \sum_{j=2}^{8} \beta_{11,j} I\left(pgpsbil_{i,t} = j\right) + \sum_{s=2}^{S} \beta_{12,s} control_sel_{i,s,t} + \varepsilon_{i,t}
\end{aligned}
$$

where $control_sel_{i,s}$ are the selected variables from our Lasso variable selection approach. (Note that here, the Lasso is run on a pooled sample ignoring the panel structure as there is no routine for running Lasso with xtreg, re. This is done to be consistent across analyses of all hypotheses.)

8. Interpretation of results:

We will focus on the estimated coefficient of having_daughter and its associated z-statistic (using robust SEs) and p-value.

- **Statistic of interest:** the coefficient estimate for the interaction of having_daughter and female, together with its z-statistic and (two-sided) p-value.
- **Interpretation:**
    - If the coefficient is positive and its two-sided p-value is below 0.05, we take that as support “in line with the hypothesis.”
    - If the coefficient is zero or negative (“not in line with our hypothesis”), or the one-sided p-value is 0.05 or higher, we conclude there is no support for Hypothesis 1.

PAP for Hypothesis 3

H3. *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).*
That is, denoting Z as a measure of gender equality in the household,
(Z for respondents with daughters) > (Z for respondents without daughters).

Formally, letting

- $Y_i = \text{dev_share_care}_i$: Percentage point deviation from equal division of caring for persons within the household, derived from pli0046 (in SOEP 2018 data), and
- $D_i = 1$ if person i has biological children and has at least one biological daughter born prior to interview ($\boldsymbol{having_daughter}_i$), 0 otherwise,

the hypothesis is

$$H1\text{: } E[Y_i|D_i = 1, \boldsymbol{X}_i] < E[Y_i|D_i = 0, \boldsymbol{X}_i],$$

where $\boldsymbol{X}_i$ denotes a set of individual-specific controls described below.

1. Variables

Main Variables

- Key Independent Variable: **having_daughter**
  - Definition: Indicator equal to 1 if the respondent has biological children and at least one biological daughter born before the 2018 interview year; 0 otherwise.
  - Construction:
    - Recode any "not applicable" code (–2) in each kidsex01…kidsex19 to valid non-missing. For each child slot j, check if kidsexj=2 (female) and kidgebj < syear. Set having_daughter = 1 if any child meets those conditions; else 0.
- Dependent Variable: **dev_share_care**
  - Derived from pli0046: "Hours Weekdays Care for Persons."
  - Construction: calculate the sum of hours spent in the entire household spent by the N members of the household. Divide individual pli0046 by sum of hours in the household. Calculate deviation from the equal division of the total hours spent (1/N) and multiply by 100.
- Control Variables
  - Measured or recoded at the 2018 interview, with any negative codes set to missing where indicated:
    - sumkids: Total number of biological children.
    - hghinc: Monthly household net income in EUR; drop observations where hghinc < 0.
    - age_at_int: Computed as 2018 – gebjahr if gebjahr > 0; otherwise missing.
    - female: Original sex variable (1 = male, 2 = female), renamed to female to state the included category in the regression.
    - erwerb: Employment status collapsed from plb0022_h:
      - Category 1 if plb0022_h=9 (not in labor force)
      - Category 2 if plb0022_h in {3,4,5,6,7,8,10,11} (non-regular employment phases)
      - Category 3 if plb0022_h=2 (part-time)
      - Category 4 if plb0022_h=1 (full-time)
    - pld0131_v2: Marital status; recoded to missing if <0.
    - migback: Migration background indicator.

        - sampreg: Geographic region of the SOEP sample (East vs. West Germany).
        - pgpsbil: Highest school-leaving qualification.
    - Optional additional covariates for LASSO Selection
        - To allow a data-driven check for omitted-variable bias, the following variables are included in the LASSO selection step (all recoded so negative values → missing):
            - corigin (respondent's country of birth, ISO numeric code)
            - fprofedu and mprofedu (father's and mother's highest professional education)
            - locchildh (place where respondent was raised up to age 15; 1=large city … 4=countryside)
            - nums and numb (numbers of sisters and brothers reported by the respondent)
            - religious affil (Religious affiliation, based on plh0258_v9 and imputed with data from surrounding years of the same variable as well as plh0258_v10 and plh0258_v11, on optional list due to potential loss of data due to sparseness in this variable for 2018)

2. Inclusion/exclusion criteria:

We include

- Survey year = 2018 only (cross-section of the SOEP 2018 wave)
- Respondents living in private households (SOEP pop codes 1 or 2)
- Respondents aged ≥ 18 at interview (computed as 2018 – gebjahr)
- Respondents with at least one biological child (sumkids > 0)
- Observations with nonmissing expressions of the dependent variable (plh0308_v2) or the key independent variable (having_daughter)
- Observations with nonmissing values on any prespecified control (rows with missing values will be dropped automatically by the regression routine)

3. Exact analysis:

We estimate the following OLS model on the dataset described with the above inclusion and exclusion criteria and variables.

$$
\begin{aligned}
dev_share_care_i &= \beta_0 + \beta_1\, having_daughter_i + \beta_2 sumkids_i + \beta_3 hghinc_i + \beta_4 female_i \\
&+ \beta_5 age_at_int_i + \sum_{j=2}^{4} \beta_{6,j} I(erwerb_i = j) + \sum_{j=2}^{5} \beta_{7,j} I(pld0131_v2_i = j) \\
&+ \beta_8 migback_i + \beta_9 I(sampreg_i = 2) + \sum_{j=2}^{8} \beta_{10,j} I(pgpsbil_i = j) \\
&+ \sum_{s=2}^{S} \beta_{11,s} control_sel_{i,s} + \varepsilon_i
\end{aligned}
$$

where $control_sel_{i,s}$ are the selected variables from our Lasso variable selection approach.

4. Interpretation of results:

We will focus on the estimated coefficient of having_daughter (β_1) and its associated t-statistic (using robust SEs, clustered on the household level) and p-value.

- **Statistic of interest:** the coefficient estimate for having_daughter, together with its t-statistic and (two-sided) p-value.
- **Interpretation:**
    - If the coefficient is negative and its two-sided p-value is below 0.05, we take that as support "in line with the hypothesis."
    - If the coefficient is zero or positive ("not in line with our hypothesis"), or the one-sided p-value is 0.05 or higher, we conclude there is no support for Hypothesis 3.

PAP for Hypothesis 4

H4. *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).*

That is, denoting Y as a measure of support for gender equality,

(Y for male respondents with daughters − *Y* for male respondents without daughters) >
(Y for female respondents with daughters − *Y* for female respondents without daughters)

Formally, letting

- $Y_i = \mathbf{gender_attitudes_i}$ index for person i (the sum of the items plh0395i01…plh0395i06 in SOEP 2019), and
- $D_i = 1$ if person i has biological children and at least one biological daughter born prior to interview ($\boldsymbol{having_daughter_i}$), 0 otherwise,
- $male_i = 1$ if a person i is male and 0 otherwise

the hypothesis is

$$H1\text{: } E[Y_i|D_i = 1, \boldsymbol{X}_i] > E[Y_i|D_i = 0, \boldsymbol{X}_i],$$

where $\boldsymbol{X}_i$ denotes a set of individual-specific controls described below.

5. Variables

- Key Independent Variable: **having_daughter, male**
  - Definition having_daughter: Indicator equal to 1 if the respondent has biological children and at least one biological daughter born before the 2019 interview year; 0 otherwise.
  - Construction:
    - Recode any "not applicable" code (–2) in each kidsex01…kidsex19 to valid non-missing. For each child slot j, check if kidsexj=2 (female) and kidgebj < syear. Set having_daughter = 1 if any child meets those conditions; else 0.
  - Definition male: Original sex variable (1 = male, 2 = female), renamed to male to state the included category in the regression. [Note that for consistency in the Do file we use the variable female, but we switch the statistical baseline for this hypothesis to match the wording of the hypothesis. Therefore, we are technically including a dummy for being male with female as the statistical baseline.]
- Dependent Variable: **gender_attitudes**
  - Items (all from 2019):
    - plh0395i01: "It's good that legislators take into account that not everyone can be classified as either male or female."
    - plh0395i02: "It should be accepted in our society that not everyone identifies with their biological sex."
    - plh0395i03: "It's completely natural that some people are born with both male and female biological features."
    - plh0395i04: "Sex assignment is not so important for a person's development."
    - plh0395i05: "Schools and preschools should promote non-gender-typical activities that are not typical for children's gender."
    - plh0395i06: "It should be accepted in our society that same-sex couples want to have children."
  - Coding and aggregation:
    - Original responses run from 1 ("Completely disagree") to 7 ("Fully agree"); negative values (<0) indicate non-response. Recode all negative values to missing. Sum the six items for respondents in 2019 to create gender_attitudes (range: 6–42).

- Control Variables
    - Measured or recoded at the 2019 interview, with any negative codes set to missing where indicated:
        - sumkids: Total number of biological children.
        - age_at_gen_att: Computed as 2019 – gebjahr if gebjahr > 0; otherwise missing.
        - erwerb: Employment status collapsed from plb0022_h:
            - Category 1 if plb0022_h=9 (not in labor force)
            - Category 2 if plb0022_h in {3,4,5,6,7,8,10,11} (non-regular employment phases)
            - Category 3 if plb0022_h=2 (part-time)
            - Category 4 if plb0022_h=1 (full-time)
        - pld0131_v2: Marital status; recoded to missing if <0.
        - plh0258_v9: Religious affiliation; recoded to missing if <0.
        - hghinc: Monthly household net income in EUR; drop observations where hghinc < 0.
        - migback: Migration background indicator.
        - sampreg: Geographic region of the SOEP sample (East vs. West Germany).
        - pgpsbil: Highest school-leaving qualification.
    - Optional additional covariates for LASSO Selection
        - To allow a data-driven check for omitted-variable bias, the following variables are included in the LASSO selection step (all recoded so negative values → missing):
            - corigin (respondent's country of birth, ISO numeric code)
            - fprofedu and mprofedu (father's and mother's highest professional education)
            - locchildh (place where respondent was raised up to age 15; 1=large city … 4=countryside)
            - hlf0135 (distance of dwelling to city centre; 1=city-centre flat … 6=≥ 60 km)
            - hlf0151 (level of worry about crime in neighbourhood; 1=not worried … 4=very worried)
            - nums and numb (numbers of sisters and brothers reported by the respondent)

6. Inclusion/exclusion criteria:

We include:

- Survey year = 2019 only (cross-section of the SOEP 2019 wave)
- Respondents living in private households (SOEP pop codes 1 or 2)
- Respondents aged ≥ 18 at interview (computed as 2019 – gebjahr)
- Respondents with at least one biological child (sumkids > 0)
- Valid household net income (hghinc ≥ 0)
- Observations with nonmissing values in the dependent variable (gender_attitudes) or the key independent variable (having_daughter)
- Observations with nonmissing values on any prespecified control (rows with missing values will be dropped automatically by the regression routine)

7. Exact analysis:

We estimate the following OLS model on the dataset described with the above inclusion and exclusion criteria and variables.

$$
\begin{aligned}
&gender_attitudes_i \\
&\quad = \beta_0 + \beta_1\, having_daughter_i + \beta_2 male_i + \beta_3\, having_daughter_i \times male_i \\
&\quad + \beta_4 sumkids_i + \beta_5\, hghinc_i + \beta_6 age_at_int_i + \sum_{j=2}^{4} \beta_{7,j} I(erwerb_i = j) \\
&\quad + \sum_{j=2}^{4} \beta_{8,j} I(pld0131_v2_i = j) + \sum_{j=2}^{5} \beta_{9,j} I(plh0258_v9_i = j) + \beta_{10} migback_i \\
&\quad + \beta_{11} I(sampreg_i = 2) + \sum_{j=2}^{8} \beta_{12,j} I(pgpsbil_i = j) + \sum_{s=2}^{S} \beta_{13,s} control_sel_{i,s} \\
&\quad + \varepsilon_i
\end{aligned}
$$

where $control_sel_{i,s}$ are the selected variables from our Lasso variable selection approach.

8. Interpretation of results:

We will focus on the estimated coefficient on the interaction between having_daughter and male (β_3) and its associated t-statistic (using robust SEs) and p-value.

- **Statistic of interest:** the coefficient estimate for the interaction of having_daughter and female, together with its t-statistic and (two-sided) p-value.
- **Interpretation:**
  - If the coefficient is positive and its two-sided p-value is below 0.05, we take that as support “in line with the hypothesis.”
  - If the coefficient is zero or negative (“not in line with our hypothesis”), or the one-sided p-value is 0.05 or higher, we conclude there is no support for Hypothesis 4.